

Supporting inequalities for polyhedra

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We consider a polyhedron described with linear inequalities:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}, \quad \text{where } A \in \mathbb{R}^{m \times n} \text{ and } b \in \mathbb{R}^m.$$

The recession cone of P is the set of directions along which it is possible to move infinitely far in P , and is given by $\text{rec}(P) = \{d \in \mathbb{R}^n : Ad \leq 0_m\}$. An inequality is *valid* for P if it is satisfied by all points in P . An inequality supports P if it is valid for P and satisfied with equality by a point in P .

Given a normal vector $\alpha \in \mathbb{R}^n$, we consider the problem of finding the right-hand-side β^* such that $\alpha^T x \leq \beta^*$ supports P :

Problem 1 (*Finding the right-hand-side of a supporting inequality*)

Given $\alpha \in \mathbb{R}^n$, what is the value β^* such that $\alpha^T x \leq \beta^*$ supports P ? ■

Throughout this note $\alpha \in \mathbb{R}^n$ denotes an arbitrary (fixed) vector. Problem 1 only has an answer under the assumption:

Assumption 1 *The polyhedron P is non-empty, and there exists β such that $\alpha^T x \leq \beta$ is valid for P .* ■

One view of β^* is that the inequality $\alpha^T x \leq \beta^*$ is valid for P , but every inequality $\alpha^T x \leq \beta$ with $\beta < \beta^*$ is *not* valid for P , *i.e.*, β^* is the supremum of the right-hand-sides of inequalities not valid for P :

$$\begin{aligned} \beta^* &= \sup\{\beta : \alpha^T x \leq \beta \text{ is not valid for } P\} \\ &= \sup\{\alpha^T x : x \in P\} \end{aligned} \quad (\text{LP})$$

The problem (LP) is a linear optimization problem, *i.e.*, a problem of optimizing a linear function with linear constraints. If Assumption 1 holds, then (LP) has a solution. Also, for any valid inequality $\alpha^T x \leq \beta$ for P , one has $\sup\{\alpha^T x : x \in P\} \leq \beta$. Hence Assumption 1 implies (LP) is bounded.

An alternative is to view β^* as the smallest right-hand-side of a valid inequality for P , *i.e.*, one has:

$$\beta^* = \inf\{\beta : \alpha^T x \leq \beta \text{ is valid for } P\}$$

The problem above can be written as a linear optimization problem by giving a linear description of the valid inequalities $\alpha^T x \leq \beta$ for P . The valid inequalities for P are characterized as follows (proved in note 2.(F)):

Fact 1 (*Characterization of the valid inequalities for P*)

For every inequality $\alpha^T x \leq \beta$ one has:

(i) *If $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$:*

$\alpha^T x \leq \beta$ is valid for $P \iff \alpha = A^T u$ and $\beta \geq b^T u$ for some $u \geq 0_m$.

(ii) *If $\alpha^T x \leq 0$ is not valid for $\text{rec}(P)$ and P is non-empty, then $\alpha^T x \leq \beta$ is not valid for P . ■*

Fact 1 shows that validity of $\alpha^T x \leq \beta$ for P depends on validity of $\alpha^T x \leq 0$ for $\text{rec}(P)$. One has that $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$ when Assumption 1 holds.

Observation 1 *If Assumption 1 holds then $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$. ■*

Indeed, if $\alpha^T x \leq 0$ is *not* valid for $\text{rec}(P)$, then Fact 1.(ii) shows that either P is empty, or $\alpha^T x \leq \beta$ is not valid for P for any β , and this is not the case when Assumption 1 holds.

It follows that under Assumption 1, Fact 1.(i) characterizes the valid inequalities for P , and this implies:

$$\begin{aligned}\beta^* &= \inf\{\beta : \alpha^T x \leq \beta \text{ is valid for } P\} \\ &= \inf\{\beta : \alpha = A^T u, \beta \geq b^T u \text{ and } u \geq 0_m\} \\ &= \inf\{b^T u : \alpha = A^T u \text{ and } u \geq 0_m\}.\end{aligned}$$

The linear optimization problem (DLP):

$$\inf\{b^T u : \alpha = A^T u \text{ and } u \geq 0_m\} \quad (\text{DLP})$$

is called the *dual* of the linear optimization problem (LP).

The problem (DLP) is feasible and bounded when Assumption 1 holds. Indeed, Assumption 1 implies *some* inequality $\alpha^T x \leq \beta$ is valid for P . Hence (DLP) is feasible. Also, Assumption 1 gives P is non-empty. This implies the value β cannot be arbitrarily small (for any $\bar{x} \in P$, all inequalities $\alpha^T x \leq \beta$ with $\beta < \alpha^T \bar{x}$ are violated by $\bar{x}$), so (DLP) is bounded.

We have now shown the important *Strong Duality Theorem*:

Theorem 1 (*Strong Duality*)

If Assumption 1 holds, then both (LP) and (DLP) are feasible and bounded, and one has:

$$\beta^* = \sup\{\alpha^T x : x \in P\} = \inf\{b^T u : \alpha = A^T u \text{ and } u \geq 0_m\}. \quad \blacksquare$$

A remaining question is how the problems (LP) and (DLP) are related when Assumption 1 is *not* satisfied. We now show that the following relationship holds when Assumption 1 fails:

Theorem 2 (*Strong duality theorem when Assumption 1 fails*)

If Assumption 1 fails, then either (a) Both problems (LP) and (DLP) are infeasible, or (b) One problem is unbounded, and the other problem is infeasible. ■

The following properties of (DLP) hold even if Assumption 1 fails:

Observation 2 (*General properties of the dual problem (DLP)*)

If (DLP) is feasible, then:

- (i) *For every $\bar{u}$ feasible for (DLP), the inequality $\alpha^T x \leq b^T \bar{u}$ is valid for P*
- (ii) *The inequality $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$ ■*

Indeed, suppose $\bar{u}$ is feasible for (DLP). Taking combinations of the inequalities $Ax \leq 0_m$ and $Ax \leq b$ with weights $\bar{u} \geq 0_m$ gives that $\bar{u}^T Ax \leq 0$ is valid for $\text{rec}(P)$, and that $\bar{u}^T Ax \leq \bar{u}^T b$ is valid for P . From $\alpha = A^T \bar{u}$ it follows that $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$, and that $\alpha^T x \leq \bar{u}^T b$ is valid for P .

We now prove Theorem 2. There are two cases, where Assumption 1 fails:

- (1) The inequality $\alpha^T x \leq \beta$ is *not* valid for P for any β :

Hence (LP) is unbounded. We will argue that (DLP) is infeasible. Suppose, for a contradiction, that $\bar{u}$ is feasible for (DLP).

Observation 2.(i) shows $\alpha^T x \leq \bar{u}^T b$ is valid for P . Hence one has $\max\{\alpha^T x : x \in P\} \leq b^T \bar{u}$, i.e., the objective value of (LP) is bounded above by $b^T \bar{u}$. However, this contradicts that (LP) is unbounded.

- (2) The polyhedron P is empty:

Hence (LP) is infeasible. For the status of (DLP), there are two cases:

- (a) $\alpha^T x \leq 0$ is *not* valid for $\text{rec}(P)$:

Here Observation 2.(ii) shows (DLP) is infeasible.

- (b) $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$:

Every inequality is valid for the empty set, so the inequality $\alpha^T x \leq t$ is valid for P for *all* t . Also, since $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$, Fact 1.(i) describes the valid inequalities for P .

Hence for every t , Fact 1.(i) shows there exists $u^t \geq 0_m$ satisfying $\alpha = A^T u^t$ and $b^T u^t \leq t$. This implies $\lim_{t \rightarrow -\infty} b^T u^t \leq \lim_{t \rightarrow -\infty} t = -\infty$, and therefore (DLP) is unbounded.